

Lecture 28: Ethics

CMSC/DATA 320: Introduction to Data Science

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Announcements

- PA7, WA7 due May 10
- Last class today (surprise!)
- Office hours continue (and end) this week!
- Course evaluations open till May 9!
- We have candy in class today (if you weren't here, YOU MISSED OUT)
- A note about the final
 - Questions similar in style to the quizzes: MCQ and fill in the blank
 - ~90 min, but we'll have 2 hours
 - You are allowed 1 side of a letter-sized piece of paper of handwritten notes
 - Blank quizzes (without solutions) will be posted later today, except for Quiz 12 (which was done on Canvas)
 - **Location: Antonov Auditorium**

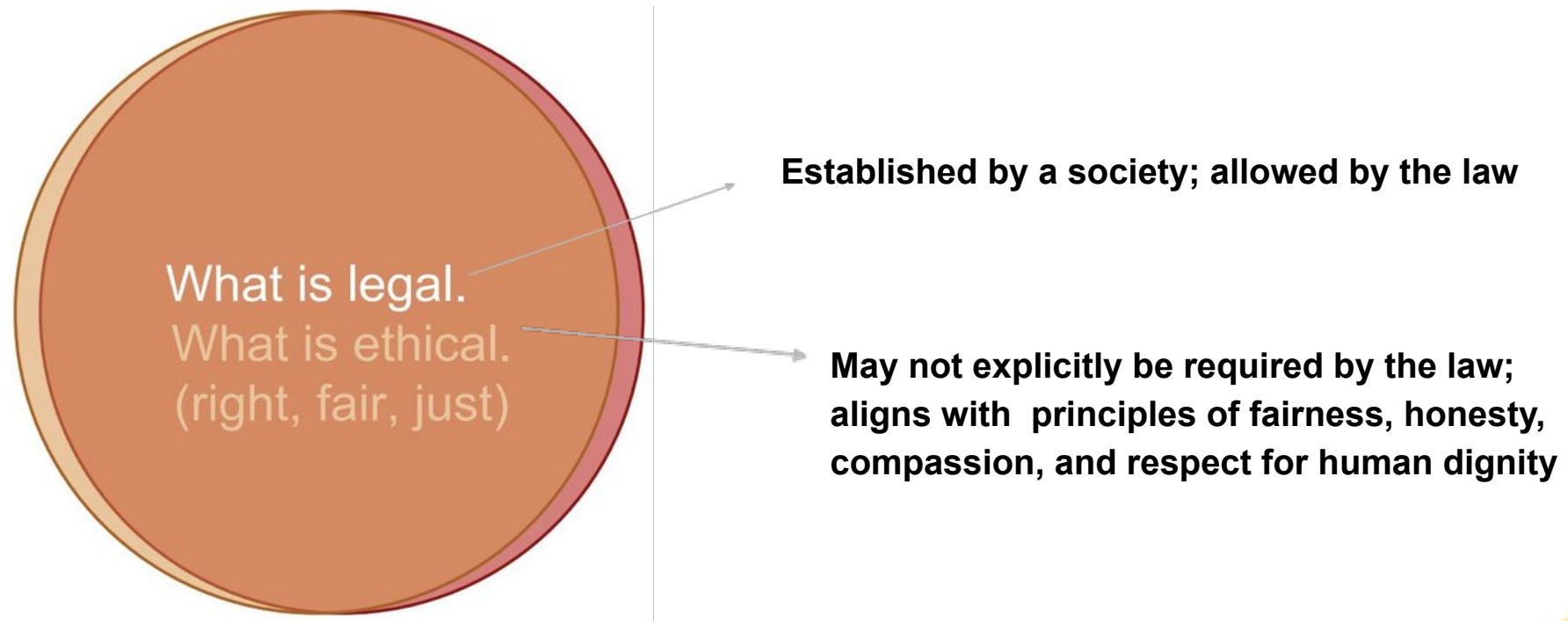
Neural Networks Demo



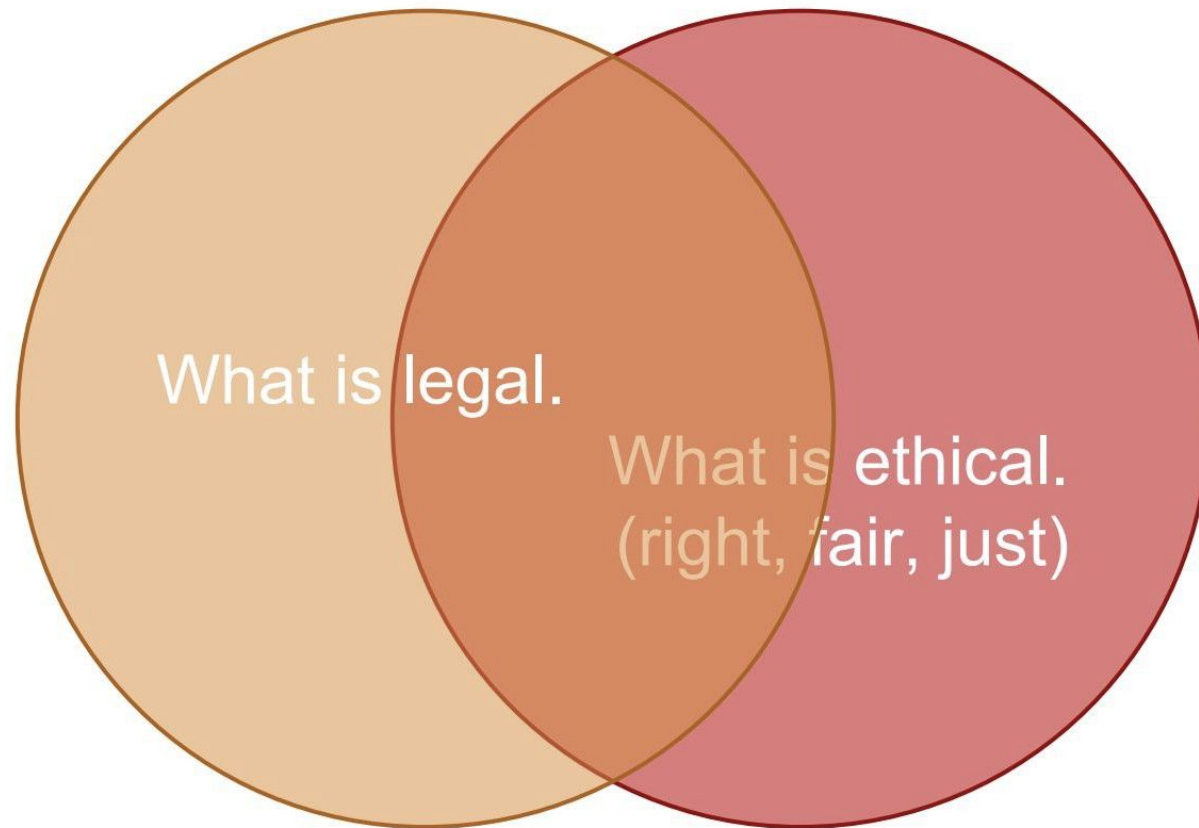
Ethics

What is right, what is fair, what is just.

In an ideal world



In the real world



We will talk about

DATA ETHICS

- A branch of ethics that evaluates data practices
 - The set of principles and processes that guide the **ethical** collection, processing, analysis, use and application of data having an effect on human lives and society
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“A better world won’t come about simply because we use data; data has its dark underside.”

— Mike Loukides, Ethics and Data Science



How Target Figured Out A Teen Girl Was Pregnant Before Her Father Did

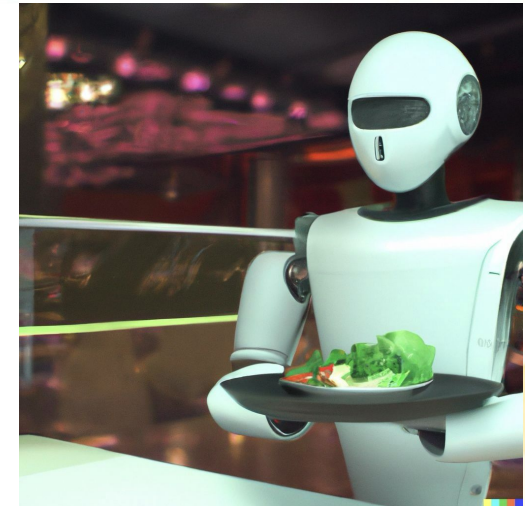
Is it Legal? Is it Ethical?

Remember, data can be powerful, yet it also has its challenges and ethical considerations. An important question that's raised is how to respect individuals' lives when using their data.

Why Ethics?

DALL-E 2 is an advanced AI capable of generating art, potentially raising questions about its impact on traditional artistic roles.

- What impacts on human artists' livelihoods? Preserving human creativity?
- How to maintain the value of artistic expression?



Case Study: Spread of harmful anti-Rohingya content

"In 2017, the Rohingya were killed, tortured...and displaced in the thousands as part of the Myanmar security forces' campaign of ethnic cleansing. In the months and years leading up to the atrocities, Facebook's algorithms were intensifying a storm of hatred against the Rohingya which contributed to real-world violence," said Agnès Callamard, Amnesty International's Secretary General."



Meta profits when Facebook users stay on the platform as long as possible, by selling more targeted advertising. The display of inflammatory content – including that which advocates hatred, constituting incitement to violence, hostility and discrimination – is an effective way of keeping people on the platform longer. As such, the promotion and amplification of this type of content is key to the surveillance-based business model of Facebook.

<https://www.amnesty.org/en/latest/news/2022/09/myanmar-face-books-systems-promoted-violence-against-rohingya-meta-owes-reparations-new-report/>

What Are Ethics?

- Behavioral norms that help society continue to function (these are empirically verifiable)
 - Personal preferences for actions or world states
 - Norms that other people expect you to adhere to
 - What a deity says to do
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Popular Personal Ethical Systems

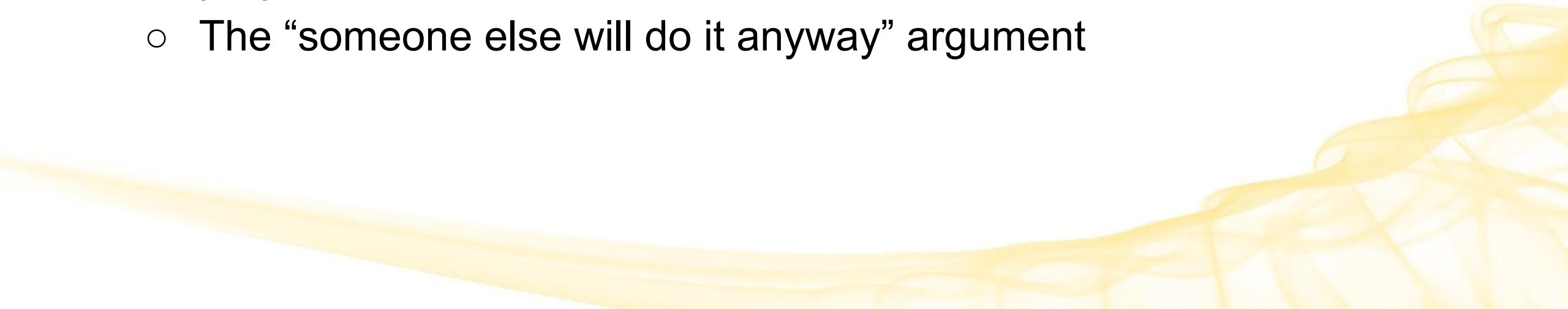
- Virtue Ethics: Having virtuous habits is good, having bad habits is bad.
 - Practicing kindness regularly vs Engaging in habitual dishonesty
- Deontology: Focus on nature of action, not consequence. Certain actions are always good, other actions are always bad.
 - Telling the truth vs Breaking a promise
- Consequentialism: Evaluate the morality of resource distribution based on the outcomes it generates. Certain world states are good, other world states are bad.
 - Policies that lead to equitable distribution vs extreme wealth inequality

If code you wrote is used to do something bad, are you responsible?



If code you wrote is used to do something bad, are you responsible?

This is a trick question!

- Will people hold you socially responsible? Maybe!
 - Should you write the code?
 - You don't necessarily know code you write will be used for nefarious purposes
 - The “someone else will do it anyway” argument
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- A decorative graphic consisting of several overlapping, flowing yellow lines that swirl and curve across the bottom right portion of the slide, creating a sense of movement and depth.

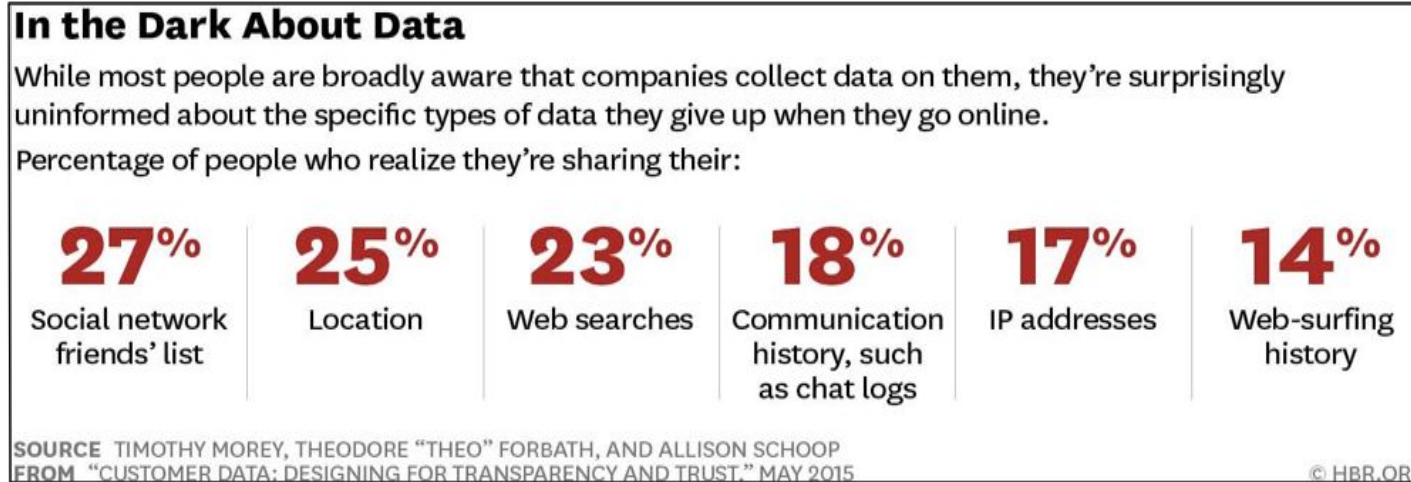
Does Data Science have impact on Ethics?



Data Science (DS) has Impact!

Provides a “better” way to do something...

But there are possible undesired consequences, for privacy, fairness, etc.



[Reference: [Harvard Business Review](#)]

- Question: How do we decide what is OK to do?
- Ethics → involves understanding the principles, values, and guidelines that govern the responsible and ethical use of data in the field of data science.
- Data Ethics education is needed...

Major Areas of Concern in Data Ethics

- Data Collection
 - Data Ownership
 - Data Privacy
 - Data Anonymity
 - Data Validity
 - Algorithm, Statistics Fairness
 - And many more...
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Data Collection

How can you be sure that your data has been ethically sourced?

“No data can be collected until explicit consent for that purpose has been given and that consent can be retracted at any time”

Informed Consent and Transparency is important!



Sampling Bias

Sampling effective at reducing the data you need to analyze

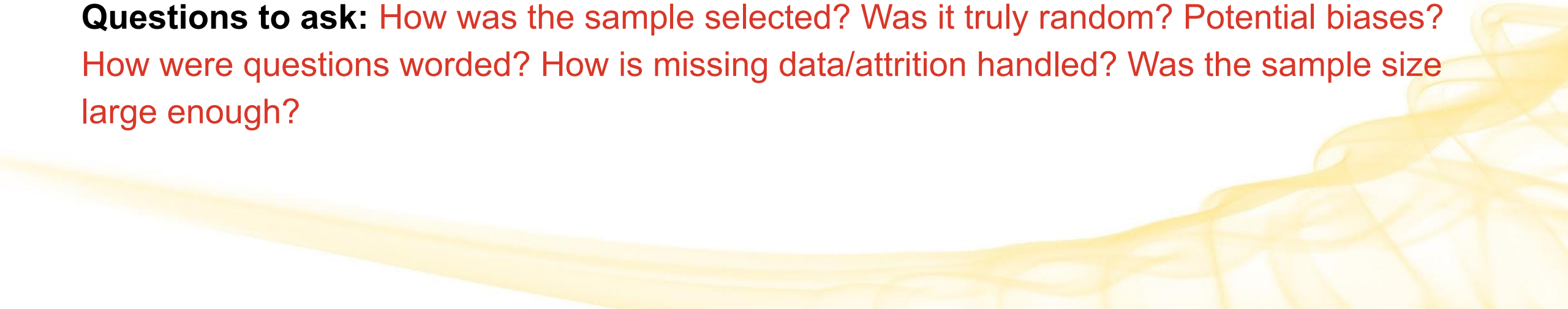
Ideally you want a random sample

- Otherwise you need to account for bias, which can be tricky

Bias in sampling: need to be very careful when generalizing inferences drawn from a sample

- Even for random samples

Questions to ask: How was the sample selected? Was it truly random? Potential biases? How were questions worded? How is missing data/attrition handled? Was the sample size large enough?



Informed Consent

- **Respect for persons** is a cornerstone value for any conception of research ethics
- Informed consent is a way to “operationalize” that principle
- Integral component of medical research for many decades
- Applicable for any research where “personal information” is divulged or human experimentation performed
- Institutional Review Boards (IRBs) in charge of implementing

How does it translate into the “big data” world?

- Much less oversight in companies

Case Study: It's Not OK,Cupid

OkCupid assigns users a “match” rating, showing how likely it thinks a pairing is to get along. The rating typically proves accurate, but, Rudder writes, “in the back of our minds, there’s always been the possibility: maybe it works just because we tell people it does. Maybe people just like each other because they think they’re supposed to?”

To test its theory, the site lied to a portion of users about how strongly they matched with other users, and observed how many single messages led to a full conversation. Sure enough, they found that two users who actually had a 90% match but were told that they had a 30% match were less likely to carry on talking than two users who actually had a 30% match but were told they had a 90% match. In other words, Rudder says, “if you have to choose only one or the other, the mere myth of compatibility works just as well as the truth.”



<https://www.theguardian.com/technology/2014/jul/29/okcupid-experiment-human-beings-dating>

Case Study: It's Not OK,Cupid

- The OKCupid experiment made people realize that there's a gap between normal website testing (A/B testing for website optimization) and conducting experiments that affect users' emotional well-being, raising questions about where the line between experimentation and fraud lies in the digital world.
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Case study: Facebook emotional experiment

Facebook routinely does A/B testing to test out new features (e.g., layouts, features, fonts, etc)

In 2014: intentionally manipulated news feeds of 700k users

- Changed the number of positive and negative stories the users saw
- Measured how the users themselves posted after that

Hypothesis: Emotions spread over the social media

Huge outcry

<https://research.facebook.com/file/2916163195290181/the-spread-of-emotion-via-facebook.pdf>

<https://www.npr.org/sections/alltechconsidered/2014/06/30/326929138/facebook-manipulates-our-moods-for-science-and-commerce-a-roundup>

Data Ownership

Who owns Data? Who has legal rights and control to make decisions about the data they generate or possess?

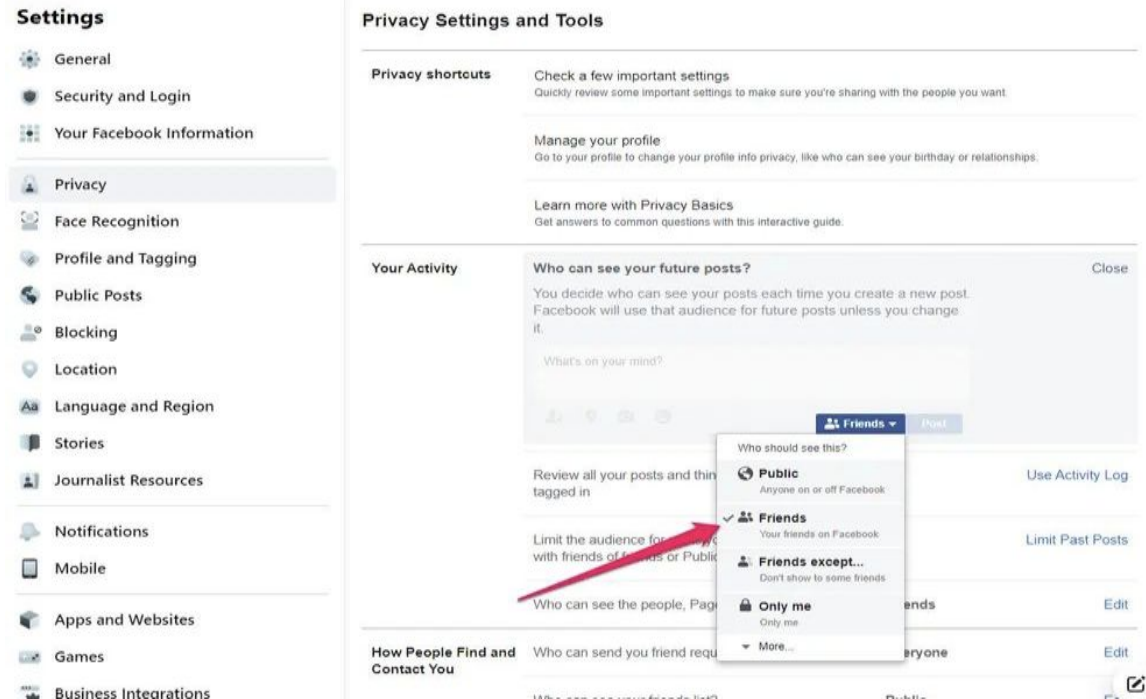
Unlawfully acquiring a photo from other person's phone without permission or consent.

Is it ethically correct?
Is this person the owner now?



Data privacy

An individual's control over their personal information's collection and use



Is it **ethical** to disclose your photo to unauthorized parties?

Is it **ethical** to use your photo without your consent?

Data anonymization

The trouble with anonymization:

For every amount of data you might release, there exists some set of priors that, if possessed by an attacker, will let them reverse your anonymization.



A survey on the patient's medical condition

Dataset (A)

Data Subject Name	Age	SSN	Medical Condition
Alice Smith	42	123-456-6666	Vision Impairment
Bob Johnson	22	223-456-7777	None
Didar Kabir	69	523-456-8888	None
Eve Jackson	44	823-456-9999	Mental Health
Gracy Singh	30	923-456-0101	Deaf or Hard of Hearing

Dataset (B)

Data Subject Name	Age	Medical Condition
18w8fy1uitycx	42	Vision Impairment
ter456kdocn	22	None
dhkj123lkdorj	69	None
ghtrxo2s034	44	Mental Health
sjjsncjdir09dos9	30	Deaf or Hard of Hearing

Which Dataset is maintaining Anonymity ?

How about Algorithmic Fairness?

A company uses an algorithm to screen job applicants for interviews



Potential Candidates		
Name	Age	Skills
John	25	...
Nick	25	...
Bob	25	...
....		...

Is the outcome of this algorithm ethically fair?

[Are Emily and Greg More Employable Than Lakisha and Jamal? A Field Experiment on Labor Market Discrimination](#) (2004) – non-AI, but AI learns on data from humans

Always Remember

It is not always simple to decide what's right or wrong.



Book recommendation

Weapons of Math Destruction: How Big Data Increases Inequality and Threatens Democracy by Cathy O'Neil

